$$dg_2 = \frac{dF}{m} = \alpha\Omega_2 = \frac{2}{3}\pi G\rho_2\Delta h, \quad (8)$$

$$g_2 = \frac{1}{3}\pi G\rho_2 a = 3.14\ m/s^2. \quad (9)$$

1.4 **[2.0점]**

U_1, v_1 .

$$\frac{mv_1^2}{2} - U_1 = 0. \quad (10)$$

$$\frac{mv_2^2}{2} - U_2 = 0, \quad (11)$$

U_2 .

: U_1 U_2.

.

.

$$U_c = 6U_1\frac{\rho_3}{\rho_2}. \quad (12)$$

, .

$\frac{a}{2}$.

$$U = G\sum\frac{m\rho_3\Delta V_i}{r_i} \sim Gm\rho_3 a^2. \quad (13)$$

4 .

$$U_c = 8\frac{U_2}{4} = 2U_2. \quad (14)$$

(12) (14)

$$U_2 = 3U_1\frac{\rho_3}{\rho_2}. \quad (15)$$

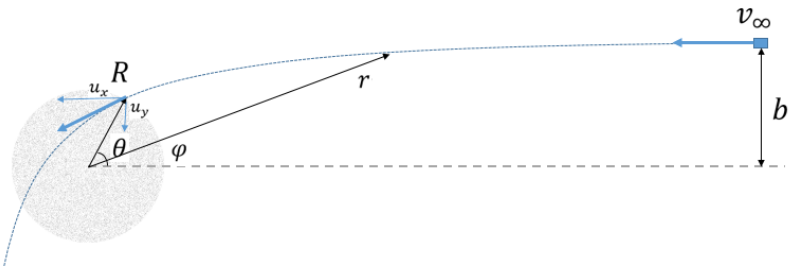
(10) (11), (15)

$$v_2 = \sqrt{\frac{3\rho_3}{\rho_2}}v_1 = 6{,}30\ km/s. \quad (16)$$

2. 먼지 구름 (6.0점)

2.1 **[2.5점]**

.



$$\frac{mv_\infty^2}{2} = \frac{mu_x^2}{2} + \frac{mu_y^2}{2} - G\frac{Mm}{R}, \quad (17)$$

$M = \frac{4}{3}\pi R^3\rho_4$.